

TECHNOVA

GUIDELINES

2026

COMPOSIT | 31ST EDITION

OVERVIEW

DESCRIPTION

Technova is a dynamic platform that celebrates innovation, research, and scientific curiosity in the field of materials science and engineering. The event invites students to present technical papers or original research work, encouraging them to explore emerging ideas, advanced technologies, and cutting-edge developments within the discipline. It provides an excellent opportunity for participants to dive deeper into technical concepts, communicate their ideas effectively, and contribute to meaningful scientific discussions.

Participants will present their work before an esteemed panel of judges and fellow students, showcasing both the depth of their research and the clarity of their presentation. The submissions will be evaluated based on originality, technical understanding, relevance, and presentation quality. Technova aims to inspire innovative thinking while giving students a chance to share their knowledge, learn from peers, and gain valuable experience in academic and technical communication.

STRUCTURE

ROUND 1 - Abstract Submission

ROUND 2 - Offline Paper Presentation

ELIGIBILITY: Open to All

TEAM SIZE: 1-3 members

FINAL EVENT DATE: 29th March, 2026

PRIZE MONEY: Rs. 10,000

RULES

RULES FOR ROUND 1

- Participants must submit an abstract (maximum 1000 words) of their paper, highlighting the objective, methodology, and key findings or insights.
- Shortlisted teams will be invited to IIT Kharagpur Campus for the final paper presentation.

RULES FOR ROUND 2

- This round will be conducted in the form of a Classical Technical conference at the IIT Kharagpur campus in front of a panel of judges,
- Participants are required to present their detailed paper within the time limit of 10 minutes, followed by a 5-minute Q&A session. Teams that exceed the given time limit will be penalised.
- There would be penalties if the time exceeds the time limit as follows -
 - 0-1 min: 10% of the presentation marks of the team
 - 1-2 mins: 20% of the presentation marks of the team
 - Greater than 2 mins: 50% of the presentation marks of the team.
- The winning team will receive cash prizes. Decisions made by the judges are final and binding.
- All participants will receive a participation certificate.

GENERAL RULES

- Team must submit only in one mentioned category/theme.
- Participants must present either their original paper or a technical paper on any topic they choose referring to existing work.
- Participants will be disqualified if any type of plagiarism is detected.

NOTE - Having a publication currently is not required, but it must be proper draft.

THEMES

- **Theme 1: Fundamentals & Extraction of Rare Earth Materials**

- Rare Earth Element (REE) classification and electronic structure
- Mining, beneficiation, and separation techniques
- Hydrometallurgical and pyrometallurgical processing of REEs
- Sustainable and green extraction methods

- **Theme 2: Rare Earths in Energy Technologies**

- Rare earth permanent magnets for wind turbines and EV motors
- REE-based materials in hydrogen production and storage
- Role of lanthanides in batteries and supercapacitors
- Rare earth oxides in solid oxide fuel cells (SOFCs)

- **Theme 3: Rare Earth Functional Materials**

- Luminescent and phosphor materials for LEDs and displays
- Magnetic materials and spintronic applications
- Optical, laser, and photonic materials based on REEs
- Dielectric, ferroelectric, and multiferroic rare earth compounds

- **Theme 4: Advanced & Emerging Applications of Rare Earths**

- Rare earths in quantum materials and advanced electronics
- REEs in thermoelectric and energy-harvesting materials
- Rare earth nanomaterials and nanocomposites
- Additive manufacturing and advanced processing of REE-based materials

- **Theme 5: Rare Earths for Sustainability & Circular Economy**

- Recycling and recovery of rare earth elements from e-waste
- Substitution and reduction strategies for critical REEs
- Life-cycle assessment of rare earth materials
- Environmental and geopolitical challenges of rare earth supply chains

THEMES & FAQ

- **Theme 6: Rare Earths in Defense, Space & Nuclear Applications**

- Radiation-resistant rare earth ceramics
- REEs in nuclear fuel and reactor materials
- Aerospace and space-grade rare earth alloys
- High-temperature and extreme-environment applications

- **Theme 7: Computational & Data-Driven Approaches**

- First-principles and DFT studies on rare earth materials
- Machine learning for discovery of REE-based compounds
- Modeling of magnetic, optical, and electronic properties
- Multiscale simulations for rare earth material design

FREQUENTLY ASKED QUESTIONS

Q1. What is the procedure for registration for the event? Do all the members have to be registered?

Yes, all the participants must first register through the COMPOSIT website, after which they will receive a participant ID. The team leader must then create a Team on the event page and will receive a Team ID.

Q2. Can a team consist of participants from different institutions?

Yes, cross-institutional teams are allowed as long as all members register and provide valid identification.

Q3. Do I need to bring my laptop with me?

Yes, you need to bring your own laptop. Make sure you have a working HDMI port or a connector with you

Q4. What is the format of the abstract to be submitted in the first round?

The abstract should be a short summary of the research paper or technical paper. This should highlight the objective, methodology, and key findings or insights of your work, all in a pdf with a maximum of 1000 words

FAQ

CONTACT

In case of any queries, contact us :

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